19. Basic chemistry for rebuilding — soap, lye, lime, charcoal, candles, vinegar, alcohol, ink and paper

Oracle OS guide. Manuals in the library: "Soap-Making Manual", "Handbook of Soap Manufacture", "Vegetable Dyes", LibreTexts Chemistry, Wikipedia. Wear gloves and eye protection and work outdoors: lye and lime burn skin and eyes (rinse 15 min with water if it happens).

Wood-ash lye (potash) — the base of soap

1. **White** ash from hardwood (not paper, plastic or barbecue briquettes) in a bucket with holes in the bottom lined with straw.
2. Pour rainwater slowly; collect the brown liquid. Run it through the ash again until strong: **an egg or potato floats** showing a coin-sized patch above the surface.
3. Boil down to concentrate if needed. Store in glass or clay (not aluminium).

Soap

- **Fat**: beef/mutton tallow, lard, olive or other oils; melt and strain.
- **Soft soap** (with ash lye = potassium): boil fat and lye, stirring for hours, until a honey-like paste with no fat on top. Test: a cooled dab should not strongly sting the tongue or feel greasy.
- **Hard soap**: add salt at the end (the paste separates and soap rises), or use bought caustic soda (weigh with a saponification table).
- Cure 4–6 weeks. Washing hands with soap prevents most diarrhoea and infections.

Lime (mortar, limewash, disinfection, nixtamal)

- Burn limestone (or shells) in a kiln at ~900 °C for 1–3 days → quicklime. **Very caustic and heats with water**: slake slowly outdoors → slaked lime (putty). Store covered with water.
- Mortar: 1 part lime putty + 3 parts sand. Limewash: lime + water like milk — disinfects and protects walls. Lime in latrines cuts flies and smell.

Charcoal

Stack dry wood, cover with earth/turf leaving vents, light from the centre; it smoulders 1–3 days with little air; when smoke turns thin and blue, seal everything and let it cool 1–2 days. Used in forges, water filters (crushed charcoal between sand) and smokeless cooking.

Candles and light

Melted tallow or beeswax; dip a cotton wick repeatedly (dipped candles) or pour into a mould. Oil lamp: olive/vegetable oil in a dish with a wick. Never leave a flame unattended.

Vinegar and preserving

Wine, cider or diluted honey fermented + vinegar "mother", in an open vessel covered with cloth, 3–6 weeks somewhere warm. Pickles vegetables and cleans.

Alcohol (antiseptic and fuel)

- Ferment fruit, honey or malted grain with yeast (or sourdough) for 1–2 weeks.
- **Distil** in a still (closed pot + copper/glass tube cooled in water): **discard the first 5% (contains methanol, toxic)**. Disinfecting skin and instruments needs 60–80% alcohol; so does burning in lamps.
- Caution: flammable vapours — keep away from open flames.

Salt

Evaporate seawater in trays in the sun or by boiling; strain through cloth first. Essential for preserving meat/fish and for livestock.

Ink, paper and glue

- **Ink**: finely ground soot/charcoal + gum arabic or glue + water; or iron-gall ink (oak galls + iron sulphate made from scrap iron in vinegar).
- **Paper**: cotton/linen rag fibres or plants (nettle, straw) boiled with ash, beaten to pulp, spread in water and lifted on a mesh frame; press and dry.
- **Glue**: boil hides, bones or hooves for hours (animal glue); cooked starch (flour paste).

Dyes

See "Textiles and leather" and "Vegetable Dyes": onion skins (yellow), walnut (brown), madder (red), indigo/woad (blue); alum or iron mordant.